

Assignment 2: Predict Diabetes using multi-layer Perceptron (MLP)

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1.Introduction

Diabetes mellitus is a worldwide chronic condition marked by high blood glucose levels. Early detection can greatly lessen complications. In this project, we developed a Multi-Layer Perceptron (MLP) neural network using the Pima Indians Diabetes dataset to classify individuals as diabetic or non-diabetic. The Perceptron serves as the basis of deep learning. Mathematically, it computes:

$$y = f(\epsilon \sum w_i x_i + b)$$

where w_i are weights, x_i are inputs, b is bias, and $f(\cdot)$ is an activation function. By stacking multiple perceptrons, MLPs can learn nonlinear relationships.

2. Dataset and Preprocessing

The dataset includes 768 instances and eight features: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age. The target variable 'Outcome' indicates diabetes (1) or non-diabetes (0). We cleaned the data by replacing zero values with medians and standardized it using StandardScaler. Then, we split the data into 80% for training and 20% for testing.

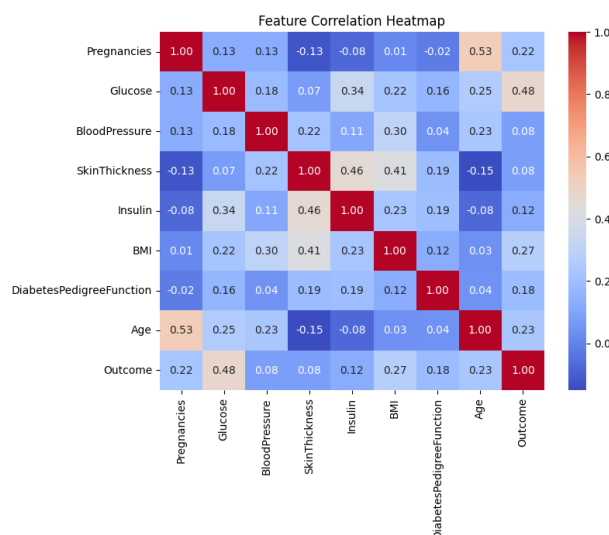


Figure 1. Feature correlation heatmap

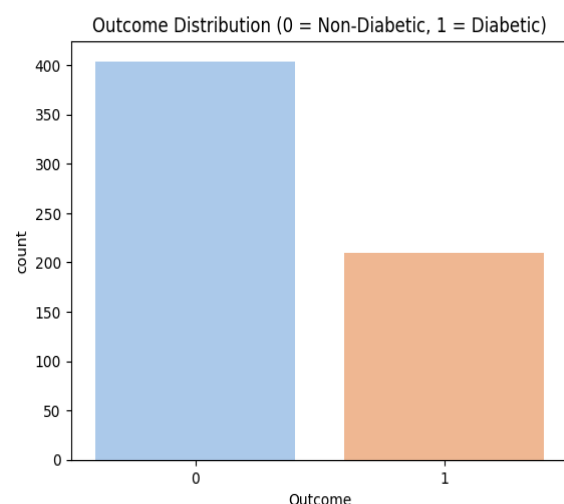


Figure 2. Outcome class distribution (imbalanced).

3. Model and Method

The MLP was built in PyTorch with two hidden layers (32 and 16 neurons), ReLU activations, and a Sigmoid output for binary classification. The Binary Cross-Entropy (BCE) loss was minimized using the Adam optimizer (learning rate = 0.001) for 100 epochs, batch size 32.

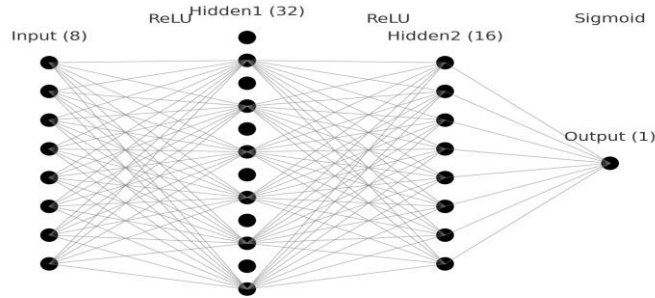


Figure 3. Multi-Layer Perceptron architecture (8->32->16->1).

4. Experiments and Results

Three experiments tested different hyperparameters to see how they affected learning stability and accuracy. Model A (32->16, lr=0.001) showed smooth convergence. Model B (16->8) underfit. Model C (lr=0.01) became unstable.

Model	Hidden Layers	Learning Rate	Accuracy	Observation
A	32->16	0.001	0.74	Stable and accurate
B	16->8	0.001	0.70	Underfitting
C	32->16	0.01	0.71	Unstable Convergence

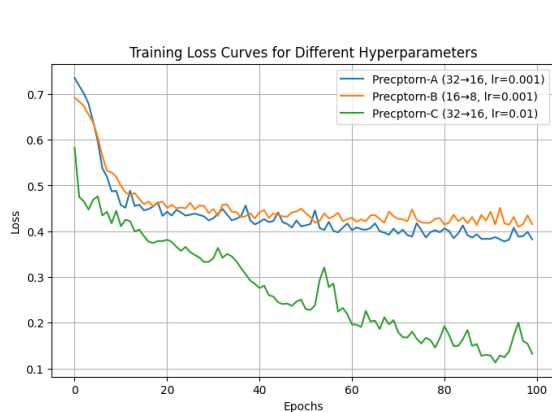


Figure 4. Training loss curve showing stable convergence.

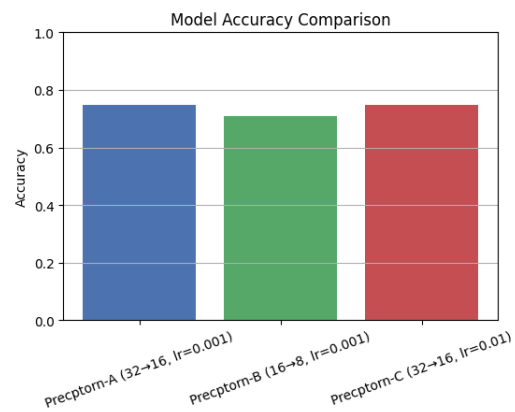


Figure 5. Accuracy comparison among three configurations

5. Evaluation and Discussion

The Binary Cross-Entropy loss decreased steadily over the epochs, showing that the gradients flowed correctly and the optimization worked well. Model A delivered the best performance, showing consistent generalization and very little overfitting. The confusion matrix shows that the model does well with non-diabetic predictions but misses some diabetic cases because of class imbalance.

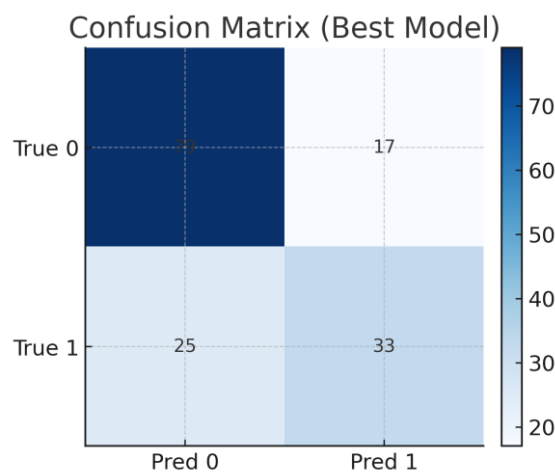


Figure 6. Confusion matrix for best model (Model A).

6. Conclusion and Future Work

The MLP achieved about 74% accuracy. This shows that simple neural networks can process structured medical data effectively. Future improvements could involve adding dropout layers, using SMOTE for class rebalancing, and applying k-fold cross-validation for better reliability.

7. References

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3. Rosenblatt, F. (1958). The Perceptron: A Probabilistic Model for Information Storage. Psychological Review.
4. University of Adelaide (2025). Assignment 2: Predict Diabetes using Perceptron/MLP.